

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO1:** Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

QUESTIONS: 13

Total Marks: 30

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **THAMIZHARASAN.S (192424087)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

THAMIZHARASAN.S

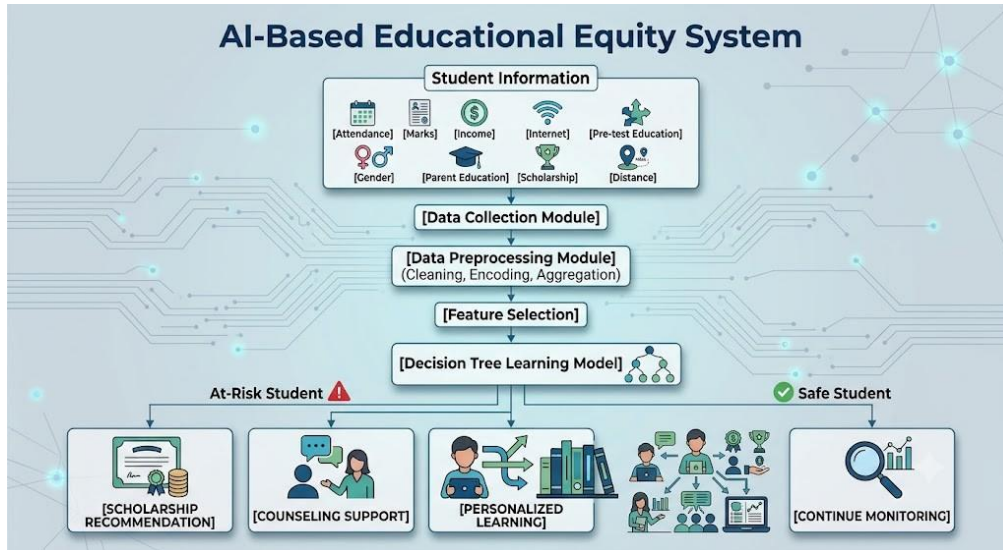
Design of an AI-Based Application to Promote Equal Educational Opportunities

Sustainable Development Goal (SDG 5): Gender Equality

Abstract

Education is a fundamental right and a key factor in achieving gender equality. However, many girls still face barriers such as poverty, discrimination, lack of internet access, and family responsibilities, which affect their educational opportunities. Artificial Intelligence (AI) and Machine Learning (ML) can help educational institutions identify students who need additional support before they drop out. This paper proposes an AI-based educational support system using the Decision Tree Learning algorithm to predict whether female students are at risk of educational inequality. The model analyzes factors such as attendance, academic performance, family income, parental education, scholarship status, and internet accessibility to classify students into "At Risk" or "Not At Risk" categories. Based on the prediction, schools can provide scholarships, counseling, mentoring, and personalized learning support. The proposed system contributes to United Nations Sustainable Development Goal 5 (Gender Equality) by promoting equal educational opportunities and improving student success through data-driven decision-making.

Keywords: Artificial Intelligence, Machine Learning, Decision Tree, Gender Equality, Educational Equity, SDG 5.



1. Problem Statement and Identification of SDG

Despite significant improvements in education, gender inequality remains a major challenge worldwide. Many girls discontinue their education due to financial difficulties, social discrimination, lack of digital resources, early marriage, and safety concerns. Educational institutions often rely on manual observation to identify students who need assistance, making early intervention difficult.

The proposed AI-based application aims to predict female students who are at risk of educational inequality using Machine Learning. By analyzing educational and socioeconomic data, the system helps schools identify students who require academic, financial, or emotional support before their performance declines.

This work supports United Nations Sustainable Development Goal 5 (Gender Equality), which focuses on eliminating discrimination against women and girls and ensuring equal access to quality education. The proposed application contributes to creating an inclusive learning environment where every student has an equal opportunity to succeed.



2. Background Study and Machine Learning Challenges

Educational institutions generate large amounts of student data, including attendance records, examination scores, family income, scholarship information, and learning behavior. Analyzing these data manually is time-consuming and often fails to detect students who require immediate support. AI-based educational systems can process these datasets efficiently, identify hidden patterns, and assist teachers in making informed decisions.

However, applying Machine Learning in education presents several challenges. Educational datasets may contain missing or inconsistent values, reducing prediction accuracy. Another challenge is data imbalance, where the number of at-risk students is much smaller than the number of regular students. Protecting student privacy and ensuring ethical use of personal information are also important considerations. Additionally, Machine Learning models must avoid bias so that predictions remain fair and do not discriminate against any group. These challenges require proper data preprocessing, balanced datasets, and transparent algorithms.

3. Learning Problem, Input Features, Target Output, and Data Representation

The proposed system uses Supervised Learning because the training data contain known outcomes. The objective is to classify students into two categories: At Risk and Not At Risk.

The input features include:

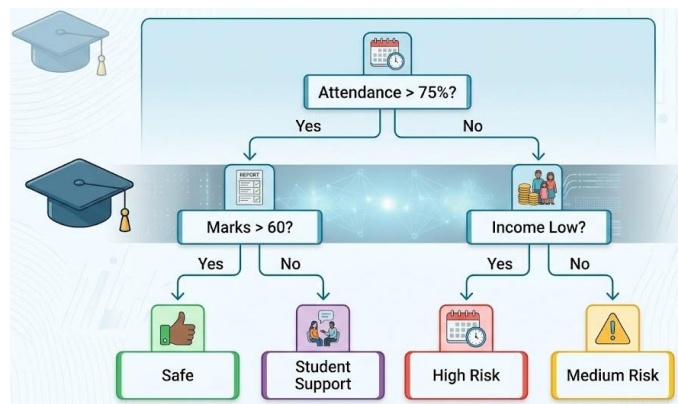
- Gender
- Attendance Percentage
- Academic Performance
- Family Income
- Parent Education Level
- Scholarship Status
- Internet Availability
- Distance to School

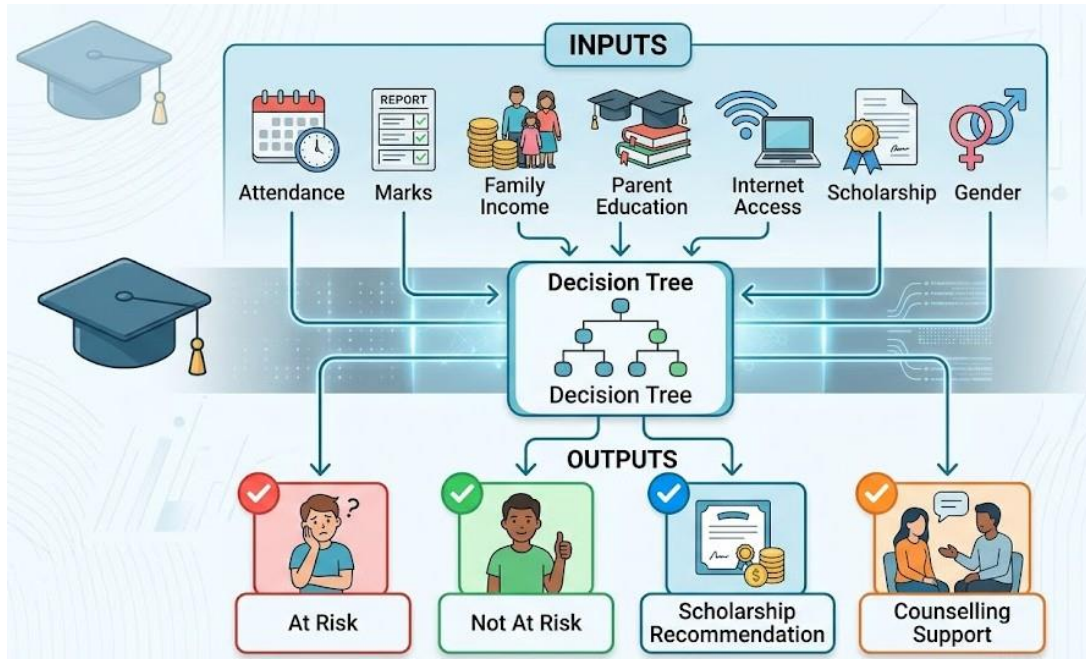
The target output is a prediction indicating whether the student requires educational support.

The data are represented in a structured table where each row corresponds to one student and each column represents an attribute.

Gender	Attendance	Income	Marks	Internet	Scholarship	Target
Female	68%	Low	58	No	No	At Risk
Female	92%	Medium	88	Yes	Yes	Not At Risk

This representation enables efficient training and prediction using Machine Learning algorithms.





4. Selection and Justification of the Machine Learning Approach

Among the Unit I Machine Learning techniques, the Decision Tree Learning algorithm is selected because it is simple, accurate, and easy to interpret. A Decision Tree classifies data by creating a sequence of decision rules based on input features. For example, the model may first evaluate attendance, followed by family income and academic performance, before predicting the student's risk level.

Decision Trees are highly suitable for educational applications because they can process both numerical and categorical data without complex preprocessing. The generated decision rules are easy for teachers and administrators to understand, making the system transparent and trustworthy. Compared with more complex algorithms, Decision Trees require less computational power while maintaining good classification performance. Their ability to explain every prediction makes them particularly valuable for educational decision-making, where fairness and accountability are essential.

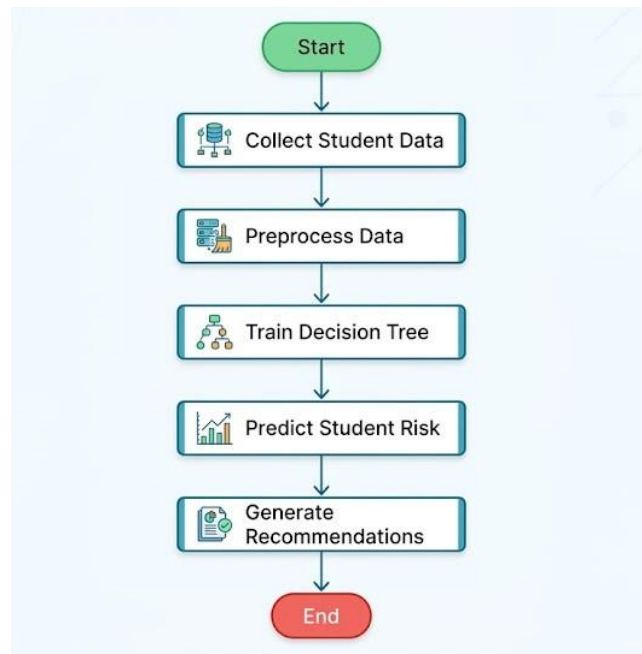
5. Proposed Algorithm and Workflow

The proposed AI application follows a systematic workflow. First, student information is collected from school records and educational databases. The data are then preprocessed to remove missing values and convert categorical attributes into machine-readable formats. The cleaned dataset is divided into training and testing datasets. The Decision Tree algorithm learns patterns from the training data and builds classification rules. During testing, the trained model predicts whether new students are at risk. Finally, the system generates recommendations such as scholarships, academic mentoring, counseling, or digital learning support.

Workflow

Student Data → Data Preprocessing → Feature Selection → Decision Tree Training → Student Classification → Educational Support Recommendation

This workflow enables educational institutions to identify vulnerable students quickly and provide timely interventions.



6. Expected Outcomes, Advantages, Limitations, and Practical Applications

The proposed AI system is expected to improve educational equality by identifying female students who require additional support before they experience academic failure or drop out. Early prediction allows schools to allocate scholarships, mentoring programs, and counseling services more effectively. The system also supports teachers by reducing manual analysis and improving decision-making accuracy.

The major advantages include fast prediction, easy interpretation, transparency, and efficient use of educational resources. Decision Trees require relatively low computational cost and generate understandable decision rules, making them practical for schools and colleges.

However, the system has certain limitations. Prediction accuracy depends on the quality of the training data. Incomplete or biased datasets may produce incorrect results. Student information must also be protected to ensure privacy and ethical use of data.

Practical applications include government education departments, schools, universities, scholarship management systems, and non-governmental organizations working to promote equal educational opportunities.

7. Recommendations and Conclusion

Educational institutions should maintain accurate student databases, regularly update Machine Learning models, and ensure that predictions remain unbiased. Student privacy must be protected, and AI should support—not replace—human decision-making.

In conclusion, the proposed AI-based educational equity system demonstrates how Decision Tree Learning can help identify students requiring educational support. By enabling timely interventions, the system promotes equal educational opportunities, reduces dropout rates, and directly contributes to Sustainable Development Goal 5: Gender Equality. The application illustrates how Machine Learning can address real-world social challenges while improving educational outcomes.

References

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3. Han, J., Kamber, M., & Pei, J. *Data Mining: Concepts and Techniques* (4th ed.). Morgan Kaufmann.
4. UNESCO. *Global Education Monitoring Report, 2023*.
5. United Nations. *Sustainable Development Goals – Goal 5: Gender Equality*.

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